ANH-DUY LE

leanhduy497@gmail.com | (+84) 349 749 495
duylebkhcm.github.io | Google Scholar

Research Interests

My research began with recognizing content and two-dimensional structure in documents, and my current document VLM work extends that foundation in multimodal understanding. In one-shot handwriting generation, I learned discrete writer-style concepts from a single reference image to guide text-conditioned diffusion, motivating my interest in composing visual concepts from few examples. I want to build generative models that use vision-language understanding to interpret a creator's intent from language, sketches, reference images, and iterative feedback. My PhD goal is personalized image and video generation with control over composition, appearance, and motion while preserving identity and temporal consistency.

Education

VNU-HCM University of Technology Aug. 2021 – Nov. 2023
Master of Engineering in Computer Science GPA: 8.29/10.00
Thesis: *A System for Extracting Mathematical Expressions from Document Images*
Thesis grade: 9.94/10.00

VNU-HCM University of Technology Aug. 2017 – Nov. 2021
Engineer of Computer Science GPA: 8.67/10.00
Thesis: *A Study on a Diagnosis System for Eye Diseases*
Thesis grade: 9.62/10.00

Publications

1. **Anh-Duy Le**, Van-Linh Pham, Thanh-Nam Vo, Xuan Toan Mai, Tuan-Anh Tran.
CONSTANT: Towards High-Quality One-Shot Handwriting Generation with Patch Contrastive Enhancement and Style-Aware Quantization.
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026. **Oral Presentation, Award Finalist.** [Paper] [Code]

2. Nam Quan Nguyen, **Anh-Duy Le**, Anh Khoa Lu, Xuan Toan Mai, Tuan-Anh Tran.
Formerge: Recover Spanning Cells in Complex Table Structure Using Transformer Network.
International Conference on Document Analysis and Recognition (ICDAR), 2023. Poster. [Paper]

3. **Anh-Duy Le**, Van-Linh Pham, Vinh-Loi Ly, Nam Quan Nguyen, Huu Thang Nguyen, Tuan-Anh Tran.
A Hybrid Vision Transformer Approach for Mathematical Expression Recognition.
International Conference on Digital Image Computing: Techniques and Applications (DICTA), 2022. **Oral Presentation.** [Paper] [Code]

Research Experience

CONSTANT: One-Shot Handwriting Generation *Apr. 2024 – Apr. 2025*
Viettel AI; advisor: Tuan-Anh Tran; first-author research

- Addressed the difficulty of recovering a new writer's style from a single, potentially noisy sample. Proposed **Style-Aware Quantization**, which learns discrete visual style concepts instead of relying on fixed high-frequency features, with a contrastive objective to separate writers.
- Introduced a **multiscale latent patch contrastive objective** that aligns generated and target patches during diffusion training, sharpening local stroke details in a single-stage framework.
- On the IAM test set, SAQ reduced FID from 16.73 to 12.47; the full model reached **FID 10.20 / HWD 0.74**, respectively **36% / 30% lower** than One-DM. Evaluated on English, Chinese, and Vietnamese handwriting.

Research Experience (continued)

Hybrid Vision Transformer for Mathematical Expression Recognition *Jun. 2022 – Jun. 2023*
HCMUT; advisor: Tuan-Anh Tran; master's research

- Addressed long-range symbol relationships and missed or repeated symbols in formula parsing using a hybrid CNN–ViT encoder with 2D positional encoding and a coverage-attention decoder; achieved **89.94 BLEU-4** and **86.48% image exact match without spaces** on IM2LATEX-100K.
- Extended the paper in my master's thesis by designing **LIMD**, a **1,564,755-pair** image–LaTeX dataset, and building its collection, normalization, rendering, and outlier-filtering pipeline. [Thesis]

Professional Experience

Viettel AI Ho Chi Minh City, Vietnam
Senior AI Engineer *Apr. 2024 – Present*
Junior AI Engineer *Oct. 2021 – Mar. 2024*

- For a Vietnamese document VLM project, built training-data pipelines: deployed a tagging system across **500,000+ real images** and produced approximately **500,000 synthetic documents** with parsing and layout annotations.
- Fine-tuned and evaluated EAGLE-3 and DFlash speculative decoders on internal document data; the best setup achieved **4.2× lower latency** than baseline Qwen3-VL-4B on B200, with unchanged internal OmniDocBench-style overall score.
- Led a **three-person team** developing text recognition models on **10M+ samples** for banking, insurance, and legal documents; achieved **98% accuracy** on an internal evaluation set covering scans and camera captures, and delivered OCR systems through production deployment.

Drimco AI Remote from Ho Chi Minh City, Vietnam
AI Engineer *Jul. 2025 – Present*

- Developed a layout analysis solution for customer technical documents, combining an M2Doc-inspired multimodal detector with rule-based post-processing; achieved **0.90 mAP** on an internal evaluation set of customer documents.
- Inspected predictions and analyzed errors in difficult layouts; experimented with fine-tuning document VLMs, including MinerU2.5 and dots.ocr, for layout analysis.

Honors & Awards

- **Award Finalist**, WACV 2026
- Runner-up, ICDAR Competition on Artistic Text Recognition 2024
- Academic Merit Scholarship (faculty GPA ranking) 2018–2019

Technical Skills

Research and modeling: Python, PyTorch, diffusion models, vision-language models.
Systems and evaluation: vLLM, MLflow, FastAPI, Docker, Git, Linux.